

KEN KANGYI PENG

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EDUCATION

PhD in Statistics

Sep 2025

Simon Fraser University, BC, Canada

Thesis: “Characterizing Stochastic Processes with Applications in Infectious Disease and Sports Analytics”

Supervisors: X. Joan Hu and Tim Swartz

MSc in Statistics

May 2022

Simon Fraser University, BC, Canada

Thesis: “Bayesian Approaches for Critical Velocity Models”

Supervisors: Tim Swartz and Gary Parker

BSc in Statistics & minor in Mathematics

May 2020

Simon Fraser University, BC, Canada

EXPERIENCE

Research

- **Postdoctoral Fellow**

Sep 2025 - Sep 2026

University Of Ottawa, Department of Civil Engineering

Supervision: Robert Delatolla

- **Research Assistant**

May 2021 - Sep 2025

Simon Fraser University, Department of Statistics and Actuarial Science, Sports Analytics Group

Project: CANSSI CRT “Sports Analytics” & CANSSI CRT “The Application of Statistical Methods to Wastewater Analysis”

Supervision: Tim Swartz and X. Joan Hu

- **Research Assistant**

Jan 2024 - Sep 2024

University Of Ottawa, Department of Civil Engineering

Project: CoVaRR-Net Wastewater Surveillance Rapid Response Research Group

Supervision: Robert Delatolla

- **Research Assistant**

Jan 2022 - May 2023

University of Waterloo, Department of Statistics and Actuarial Science

Projects: MfPH - NSERC “Emerging Infectious Diseases Modelling Initiative”

Supervision: Charmaine Dean

- **Research Assistant**

Mar 2020 - May 2021

Simon Fraser University, Department of Biomedical Physiology and Kinesiology

Supervision: Tim Swartz and David C. Clarke

Industry and Application Research Experience

- **Co-founder & Chief Data Scientist**

Jun 2023 - Current

AccMov Health Inc.

Lead data strategy, collaborate with sports and health scientists to analyze data and drive business insights

- **Sports Analyst**

Jan 2022 - Dec 2023

Stathletes

Hockey analytics

- **Co-op Data Scientist** *Oct 2021 - Dec 2021*
Public Health Agency of Canada
Worked with database of ArriveCAN App
- **Co-op Statistician** *May 2019 - Sep 2019*
BC Centre for Excellence in HIV/AIDS
Worked with health administrative data
- **Co-op Data Scientist** *May 2018 - Dec 2018*
Statistics Canada
Worked with Canada's multifactor productivity database & T2 leap linked administrative database

TEACHING EXPERIENCE

- **Instructor**
Simon Fraser University
STAT 445/645: Applied Multivariate Analysis *Spring 2026*
- **Guest Lecture**
Simon Fraser University
STAT 240: Lecture on text data manipulation *Spring 2025*
- **Teaching Assistant**
Simon Fraser University
STAT 240: Introduction to Data Science *Spring 2025*
STAT 302: Analysis of Experimental and Observational Data *Fall 2023*
STAT 830: Statistical Theory I *Fall 2022*
STAT 261: Laboratory for Introductory R for Data Science *Fall 2021*
STAT 201: Statistics for the Life Sciences *Summer 2021*
STAT 380: Introduction to Stochastic Processes *Spring 2021*
STAT 203: Introduction to Statistics for the Social Sciences *Fall 2020*
STAT 100: Chance and Data Analysis *Fall 2020*
- **Informal Mentoring**
Julian Lau — Undergrad (UAlberta), with Profs. Bonita Lee & Lilly Pang *Aug 2025 – Present*
Krystal He — MSc (SFU), with Prof. Joan Hu *May 2025 – Present*
Riley Isaac — Undergrad & MSc (SFU), with Prof. Joan Hu *May 2024 – Present*

HONORS AND AWARDS

CANSSI Distinguished Postdoctoral Fellow – \$140,000 CAD	<i>2026–2028</i>
Multiple Graduate Fellowships, SFU – Total: \$17,500 CAD	<i>2021–2025</i>
Multiple PhD Research Scholarships, SFU – Total: \$9,122 CAD	<i>2023–2025</i>
SFU Travel and Research Award – \$2,054 CAD	<i>2024–2025</i>
MfPH – Career Networking & Industry Partnership Event Travel Funding – \$1,500 CAD	<i>Fall 2023</i>
Mathematics for Public Health Festival Travel Funding – \$2,000 CAD	<i>Fall 2022</i>
Dr. Gord Sleivert Young Investigator Award – \$500 CAD	<i>Summer 2020</i>
NSERC Undergraduate Student Research Award – \$4,500 CAD	<i>Summer 2019</i>

RESEARCH PAPERS

Methodological Research

1. **Peng, K. K.**, Dean, C. B., Delatolla, R., & Hu, X. J. “Learning associations of COVID-19 hospitalizations with wastewater viral signals by Markov modulated models.” *Epidemics*: 100840 (2025).
2. **Peng, K. K.**, Delatolla, R., Hu, X. J., Renouf, E. & Dean C. B. “Joint Modeling of Two Stochastic Processes, with Application to Learning Hospitalization Dynamics from Wastewater Viral Concentrations.” *Canadian Journal of Statistics*, in press. Available at: <https://arxiv.org/abs/2601.07977>.
3. **Peng, K. K.**, Hu, X. J. & Swartz, T. B. “Modeling Event Dynamics by Self-Exciting Processes with Random Memory.” Submitted. Available at: <https://arxiv.org/abs/2601.07980>.
4. **Peng, K. K.**, Guan, T., & Zhao, X. “Hierarchical Bayesian Joint Modeling of Athlete Performance Trajectories and Latent Peak Ages.” Submitted.

Applied Research

5. Wen, J., **Peng, K. K.**, et al. “Site-specific wastewater-based surveillance in early detection of COVID-19 new cases and prediction of mass testing outcomes in long-term care facilities.” *Scientific Reports*, 16, Article 17235 (2026). Available at: <https://www.nature.com/articles/s41598-026-47833-8>.
6. Hegazy, N., **Peng, K. K.**, et al. “Twelve distinct laboratory methods used to measure SARS-CoV-2 in wastewaters throughout a three-year Ontario-wide, Canada study: Impact on public health interpretation of disease incidence.” *Journal of Water Process Engineering* (2026). Available at: <https://www.sciencedirect.com/science/article/pii/S2214714426008238>.
7. Isaacs, R. L.^{*}, Hu, X. J., **Peng, K. K.**, & Swartz, T. B. “Learning About Corner Kicks in Soccer by Analysis of Event Times Using a Frailty Model.” In *Sports Analytics (ISACE 2026)*, Lecture Notes in Computer Science, 16610, 442–455 (2026). Available at: https://link.springer.com/chapter/10.1007/978-3-032-27272-0_28.
8. Hegazy, Nada, **Peng, K. K.**, et al. “Variability of Clinical Metrics in small population communities drive perceived wastewater and environmental surveillance data quality: Ontario, Canada-wide study.” *ACS ES&T Water* (2025).
9. **Peng, K. K.**, Hu, X. J. & Swartz, T. B. “On the time of corner kicks in soccer: an analysis of event history data.” *Computational Statistics* (2024).
10. **Peng, K. K.**, Brodie, R. T., Swartz, T. B., & Clarke, D. C. “Bayesian inference of the impulse-response model of athlete training and performance.” *International Journal of Performance Analysis in Sport*, 1-16 (2023).
11. **Peng, K. K.**, Renouf, E. M., Dean, C. B., Hu, X. J., Delatolla, R., & Manuel, D. G. “An exploration of the relationship between wastewater viral signals and COVID-19 hospitalizations in Ottawa, Canada.” *Infectious Disease Modelling* (2023).
12. **Peng, K. K.**, Clarke, D. C., Swartz, T. B. “Bayesian approaches for critical velocity modelling of data from intermittent efforts”. *International Journal of Sports Science & Coaching* (2022).
13. Zhao, X., **Peng, K. K.**, Cournoyer, J., & Graham, R. B. “Seasonal changes in neuromuscular and biomechanical performance in female soccer players: Implications for athlete monitoring.” *International Journal of Sports Science & Coaching* (2026). Available at: <https://doi.org/10.1177/17479541261479187>.

^{*}Mentored student.

PRESENTATIONS

Invited Presentations

- 2026 The 7th ICSA-Canada Chapter Symposium: Applications of functional data, time series data, and stochastic processes
Montreal, QC, Canada
- 2026 Conference on Econometrics and Statistics: Sports analytics (Special Invited Session)
Kyoto, Japan
- 2026 Centre de recherches mathématiques (CRM): Invited presentation (joint with Professor Charmaine Dean)
Montreal, QC, Canada
- 2026 WNAR of the International Biometric Society: Novel statistical methods for complex event history data
Pullman, Washington, US
- 2026 Department of Mathematics and Statistics Seminar, Wuhan University
Wuhan, China
- 2025 Conference on Econometrics and Statistics: Infectious disease modeling
Tokyo, Japan
- 2025 International Conference on Statistics and Data Science: Environmental modelling
Vancouver, BC, Canada
- 2025 WNAR of the International Biometric Society: Novel models of survival data
Whistler, BC, Canada
- 2025 Lifetime Data Science: Recent advances in life history data analysis with multi-state models
Brooklyn, NY, US
- 2024 Joint Statistical Meetings: Newly developed point process models for repeated events
Portland, Oregon, US
- 2024 WNAR of the International Biometric Society of the International Biometric Society: Sparse modeling for biomedical data
Fort Collins, Colorado, US
- 2023 WNAR of The International Biometric Society: Statistics in biosciences
Anchorage, Alaska, US
- 2022 BIRS-CMO: Statistical challenges in the identification, validation, and use of surrogate markers
Banff International Research Station. Remote

Other Presentations

- 2026 iMicroSeq meeting: Methods evaluation to enable future data sharing and integration
Scheduled, Remote
- 2023 Fields MfPH - Career Networking and Industry Partnership Event, The Fields Institute
Toronto, ON, Canada
- 2023 North America Machine Learning, Optimization and Statistics Symposium, poster presentation
Vancouver, BC, Canada
- 2022-2023 Colloquium on Mathematics for Public Health, The Fields Institute
Remote

- 2022 Mathematics for Public Health Festival (MfPHest), Poster presentation, The Fields Institute
Toronto, ON, Canada
- 2021 Canadian Statistical Sciences Institute collaborative research teams showcase event
Remote
- 2021 Statistical Society of Canada annual meeting, Poster presentation
Remote
- 2020 The SPort INnovation Summit, recipient of the Dr. Gord Sleivert Young Investigator Award
Remote

ACADEMIA ACTIVITIES

Leadership

- **Chair**, Invited session at 2026 ICSA: *Recent Statistical Developments Stimulated by COVID-19 Pandemic* 2026
- **Organizer & Chair**, Invited session at 2026 EcoSta: *Statistical modeling for environmental surveillance* 2026
- **Co-organizer**, BIRS 5-day workshop: *Advances in Statistical Methodology for the Analysis of Wastewater Data* 2026
- **Co-organizer & Chair**, Invited session at 2025 EcoSta: *Innovative statistical approaches in wastewater-based epidemiology: Recent developments and applications* 2025
- **Lab Meeting Coordinator**, Dr. Joan Hu's Lab 2024–2025

Paper Referee

- Statistics in Medicine
- Computational Statistics
- Canadian Journal of Statistics
- Statistical Theory and Related Fields
- BMC Medical Research Methodology
- Journal of Hospital Infection

Others

- **Committee Member**, Student and Recent Graduate Committee (SARGC), Statistical Society of Canada 2026–2027
- **Review Committee Member**, 2026 WNAR Student Paper Competition 2026
- **Volunteer**, Vancouver, International Conference on Statistics and Data Science 2025
- **Program Assistant**, Vancouver, Cascadia Symposium on Statistics in Sports 2023–2024
- **Volunteer**, Vancouver, North America Machine Learning, Optimization and Statistics Symposium *Summer 2023*
- **Next Generation Member**, The Fields Institute for Research in Mathematical Sciences, Mathematics for Public Health (MfPH) program 2022–2024